

Penguins on wheels

Part number 8527



an NXT as a control unit

Polulu MP 12 V 9.7:1



Pololu-motors for propulsion and dribbler

Our robot

LEGO Ultrasound



Ultrasonic sensor for orientation

IRSeeker V2 sensor

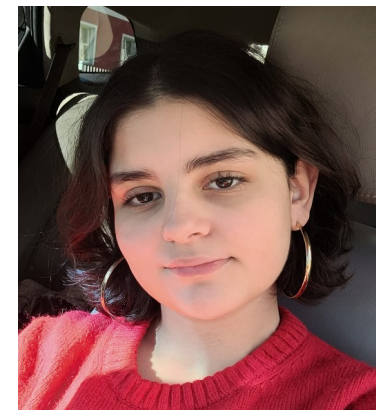
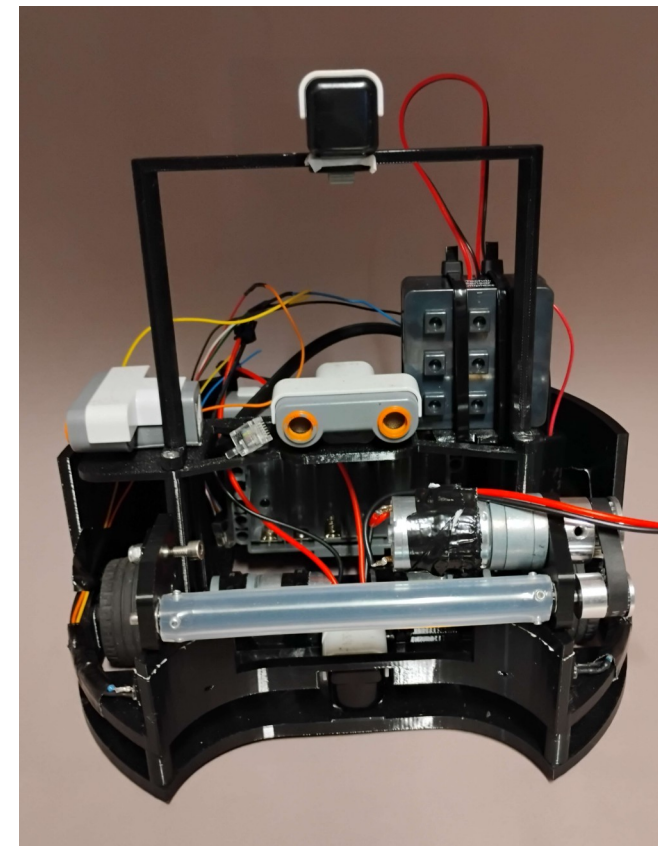


A seeker for ball detection

HiTechnic compass sensor



A compass for orientation



Thalia programmed the robot (13.Y)



Justus designed the robot. (16.Y)



A light barrier to detect whether the ball is located in the Capturing Zone.

HiTechnic Sensor Multiplexer



A multiplexer for using three ultrasonic sensors

At the beginning of the game, the playing direction is manually calibrated using the goals. After kickoff, the robot drives toward the ball at full speed. Using the light barrier sensor, it detects whether it has possession of the ball or not. Depending on this, it either drives toward the goal using the compass or follows the infrared emitting ball with the infrared seeker.

While moving around the field, the robot is able to determine its position. With the ultrasonic sensor, it checks whether it is about to collide with a wall when the measured value becomes small. If this happens, the robot drives backward.

If the robot can no longer detect the ball, it returns to its own goal and defends it until the ball is visible again, after which it resumes tracking and following the ball.